

Alexandre Milharado

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SUMMARY

Recent Master's graduate who ranked among the **top 5% of students throughout Bachelor's and Master's studies**, with practical experience gained through a summer internship and teaching assistant roles for two university courses. Specializes in **Software Engineering** and **Machine Learning**, with expertise in developing scalable applications, data-driven solutions, and intelligent systems. Demonstrates strong analytical thinking and structured problem-solving skills, with the ability to decompose complex challenges into manageable tasks and deliver effective solutions.

EXPERIENCE

Teaching Assistant - Part-time

ISCTE - University Institute of Lisbon

September 2025 - February 2026, Lisbon, Portugal · On site

- Taught **Introduction to Programming and Concurrent and Parallel Programming**.
- Evaluated **more than 30 student project groups** and graded **more than 500 exam questions**.

Full Stack Developer - Summer Internship

Bubble-go (Acquired by Sword Health)

July 2023 - September 2023, Lisbon, Portugal · Hybrid

- Developed an event management application, designing and implementing **more than 5 screens** in **React Native**.
- Built **10 server-side functionalities** for request processing using **NestJS** and **MongoDB**.

PROJECTS

Image Reconstruction from Event Cameras | Python & Pytorch

ISCTE - University Institute of Lisbon · github.com/AlexandreMilharado/JAE2VID · February 2026 - September 2026

- Studied CCD/CMOS technologies, sensor noise models, biological vision systems, and event-based sensors (DVS, ATIS, DAVIS).
- Implemented and benchmarked reconstruction approaches, including E2VID, analyzing their performance on event-stream data and image quality metrics.

Autonomous Navigation | Python & Webots

ISCTE - University Institute of Lisbon · github.com/AlexandreMilharado/Robotics · April 2025 - June 2025

- Designed a deep neural network trained via genetic algorithms for obstacle avoidance and path following in a closed arena.
- Trained a Thymio robot using PPO and RPPO to explore environments while avoiding obstacles and ledges.
- Deployed trained models from simulation to a real-world Thymio robot for autonomous exploration and obstacle avoidance.

JSON Library | Kotlin

ISCTE - University Institute of Lisbon · github.com/pafmoura/KJSON · April 2025 - May 2025

- Developed advanced operations such as filtering, mapping, custom visitors and serialization.
- Implemented automatic conversion of data classes, primitives, collections, enums, and maps into JSON structures using reflection.
- Added an HTTP server framework with annotation-based with automatic JSON serialization and path/query parameter support.

MongoDB to MySQL Migration via MQTT | Java, MongoDB & MySQL

ISCTE - University Institute of Lisbon · github.com/K3vinb5/Projeto_PISID_Grupo_12 · February 2024 - May 2024

- Designed schemas for both MongoDB and MySQL to support efficient, consistent data migration.
- Built a real-time data streamer using MongoDB replicas to sync inserts into MySQL with minimal data loss.
- Implemented data operations in Java to ensure data integrity and usability during migration.

TripAdvisor Replica for Spot Discovery & Social Interaction | Django, jQuery & React

ISCTE - University Institute of Lisbon · github.com/AlexandreMilharado/Spot-Discovery · February 2024 - May 2024

- Built dynamic page interactions using jQuery to enable editing, actions, and infinite scroll without full-page reloads.
- Implemented server-side filtering (e.g. bad word detection).
- Developed REST API endpoints with token-based authentication for Ajax & React integration.

EDUCATION

Master of Science in Computer Engineering

ISCTE - University Institute of Lisbon · Lisbon, Portugal · 2026 · Grade: 18.5/20.0

- Consistently ranked among the **top 5% of students** in the Computer Engineering program each academic year.

- Led **2 JavaScript workshops** for incoming students, providing hands-on coding experience.
- Elected **Class Representative**, representing the Computer Engineering student body.

Bachelor of Science in Computer Engineering

ISCTE - University Institute of Lisbon · Lisbon, Portugal · 2024 · Grade: 17.3/20.0

- Consistently ranked among the **top 5% of students** in the Computer Engineering program each academic year.
- Volunteered as a **Student Ambassador**, led **2 JavaScript workshops** for incoming students, providing hands-on coding experience.
- Elected **Member of the Pedagogical Council**, representing the Computer Engineering student body.

CERTIFICATIONS

Microsoft Azure Fundamentals AZ-900

Microsoft · 2026

Winning Team of the Mathematics and Programming Competition Certificate - Twice

FISTA · (2023 & 2024)

More than 5 certificates in Data Science

Kaggle - Data Science Company · 2024

More than 5 certificates in Transversal Skills

ISCTE - University Institute of Lisbon · 2024

Cambridge English B2 First – Score 172

Cambridge University Press & Assessment English · 2022

SKILLS

Industry Knowledge: Computer Networks, Artificial Intelligence (AI), Data Science, Cybersecurity, Object-Oriented Programming (OOP), Imperative Programming, Database, Operating Systems, Scrum

Tools & Technologies: Kotlin, Scala, Java, Python, React Native, TypeScript, JavaScript, HTML5, CSS, Postman API, Bash, C, MySQL, MongoDB, Microsoft Excel